

Art vs. Science: What Exactly is Mathematics?

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December 10, 2021

Abstract

This not-so-long paper will primarily discuss the origin of mathematics; how mathematicians define it through long considerations and debates; and, most importantly, how it relates to the concept of art and science. The reading fortunately requires almost no mathematical background and solely focuses on the philosophical standpoint of the journey of mathematics. Special thanks are dedicated to dear mathematician friends who have contributed to this essay by sharing thoughts and feedbacks during the making.

When the word ‘mathematics’ shows up in any kind of context, the first thought that might come in many people’s minds is some stuff to do with numbers. If that is indeed the case, then that would make math so dull and colorless, leaving out a huge range of other topics like geometry and topology. This illustrates how the majority of people are unaware of what mathematics truly means and what its essence is. Most certainly, there is more to math than meets the eye. Illustrious scholars have debated this matter until they were blue in the face, and yet no consensus has been reached about whether mathematics is a natural science, a branch of the humanities, or an art form.^[1]

1 Before Definitions Make Their Way, an Axiom was Agreed Upon

Even though it might seem impossible to define what math is, the fact that it is still considered as the most essential subject to be taught at educational institutions and the extensive work of mathematicians in developing this discipline suggest that there could be a glimpse of hope to find the answer to this very question. The most intuitive way to approach the definition of mathematics is to examine which things lie within the scope of mathematics and which are not.

1.1 The Existence of Infinitely Many Primes

Various events are happening at the start of a day: breakfast, catching the bus, going to school, etc. But the first thing that most people are engaged with is the alarm clock, and checking the time often becomes another habit before doing any other activities in the morning. In fact, humans are so comfortable with those numbers that the standard concept of time is hardly realized to be related to mathematics. People’s daily life are



Figure 1: Photo by Reuben Teo on Unsplash

strictly based on the 24-hour clock and other time conventions adopted, which are deeply interconnected with an area of math called number theory.

Now, what about the music people listen to from their headphones or radio on their way to the office? Those bouncy beats and consistent rhythm that form a beautiful masterpiece seem like a work of magic. Despite its complexity, who would ever have thought that the harmonic scale of music is highly related to an amazing area of abstract algebra called group theory!

What about the money that people attempt to manage and use to buy tickets, which are highly based on the mathematical optimization problem? Or the expectation of a rain to pour down today that can be modeled perfectly with probability theory? Or even chairs that people use to sit in front of their screens for hours, which are designed to be comfortable because of something so fundamental as descriptive geometry?

These examples of mathematics occurrence revolve around the inventions of mankind. But what about the nature itself? Well, the famous golden ratio can be easily found in flower patterns, the shape of seashells, or the spiraling galaxies. The term ‘golden’ represents geometrical perfection, that it is even adapted to architecture in designing buildings and breathtaking staircases!

Previous instances highly suggest that nearly everything in the universe is somehow connected to mathematics. However, many great thinkers would say that it is too naive to jump directly into conclusions about mathematics being the universal language of everything, especially not without proofs. At the same time, it is also exceedingly difficult to list out things that lie outside the scope of math. Consider anything nearby, and it will most likely find its way to be interconnected with the mathematical world, either through physics or any other subjects of study.

1.2 A Unique Line Passing Through Two Distinct Points

As much as it is tempting to say that previous examples, such as time and money, really revolve around mathematics, it is nothing but a mere claim without any basis for reasoning. That is because there is nothing to agree on in the first place.

Consider the following thought experiment: Two best friends walk into some restaurant to have lunch together. The diner provides a huge range of menus, including their favorites. However, as in some circumstances, they have polar opposite preferences. One of them prefers soup-based ramen, while the other says dry ramen is the best among the

other menus. It is certainly impossible to conclude who has the right opinion, simply because everybody has different taste. The question is, how come they still strangely enjoy having meals together, despite the huge gap of preferences that potentially lead them into some debates?

This sort of phenomenon occurs almost everywhere. For instance, a group of students with different backgrounds may work on a single project together without any conflict. Likewise, a group of people who are practically strangers to one another may assemble together to form a cohesive basketball team.

The most fundamental answer to this is simply because people have what is called empathy. This unique ability of humans to try to accept others' definitions of what is good and what is bad is the one that brings them together.[2] The mutual understanding enables people to create ideal bonds and discover new things. Empathy makes it possible to have agreement on something, and this is exactly how mathematics grows!

People can agree on the fact that $1 + 1 = 2$ simply because they are willing to accept the fact that numbers behave like so. People can have something to talk about simply because those things are believed to exist. In order to discuss about mathematics, a set of rules, namely axioms, has to be agreed upon in the first place. Any theory or definition would fall off if there was no grounded basis assumed. Although something is true, if no two people agreed that it is true, there is no point in discussing it in the first place.

Mathematicians from all over the world try to accept and understand the difference between one idea and another; then they attempt to connect all those imaginations together in a single, solid agreement that erases all those differences. And just from there, they are able to play and discover so much more with a universal language that everybody can understand. And that's when mathematics was truly born.

1.3 Independent Vectors that Generate the Whole Space

The birth of mathematics raises a big philosophical question about its origin: Was mathematics discovered or invented?

There are undoubtedly huge debates between these two viewpoints. On one side, humans observed nature, and then tried to describe it as accurately as possible. As a result, they had to invent some kind of instrument that helped them build a better understanding of the nature of reality. And that is when mathematics was invented. This view would imply that since the very beginning, all physical objects lived independently without mathematics. The extreme usefulness of mathematics in describing how the universe works is just a mere happy coincidence. Hypothetically speaking, if there were another civilization out there that was intelligent enough to describe the world, they would most likely invent a completely different piece of mathematics!

On the other hand, it is the observation of reality that leads humans into the world of mathematics. The natural and social phenomenon serves as a clue to unlock a deeper humongous realm that is completely separated from the tangible world. And so with the creativity and intelligence of humanity, mathematics is discovered and gradually uncovered by great detectives called mathematicians. This view would mean that since the very beginning, mathematics already exists solely even without the existence of nature. Therefore, if another new math was born, then it would certainly have been embedded within the existing math, and even brought no novelty to the current knowledge!

The question about whether mathematics was discovered or invented can be boiled down into whether mathematics is a manmade artifact or a natural kind. History, sur-

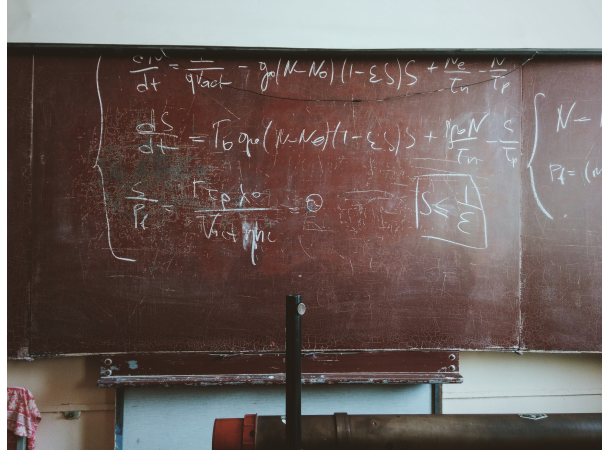


Figure 2: Photo by Roman Mager on Unsplash

prisingly, demonstrates that mathematics is a story of discovery rather than invention. Although it was human curiosity and intuition that recognized the underlying principles of mathematics, and human ingenuity that later provided various means of recording and notating them, those principles themselves are not a human invention. The fact that $2 + 2 = 4$ is true, independent of human existence; the rules of mathematics, like the laws of physics, are universal, eternal, and unchanging. When mathematicians first showed that the angles of any triangle in a flat plane when added together come to 180° , a straight line, this was not their invention: they had simply discovered a fact that had always been (and will always be) true.[3]

However, the existence of mathematics undeniably depends on the set of axioms established. Some recent branches of mathematics—such as Group Theory and Logical Calculus—were invented along with all the definitions and notions within them. Even when Calculus was believed to be invented by Newton and Leibniz independently, the notations they used were totally distinct from one another. Interestingly, yet, all the ideas that came together converged into a same exact conclusion that mathematicians are using nowadays![4]

To that end, mathematics was really both discovered and invented. The boundary between artifacts and natural kinds is essentially fuzzy when talking about mathematics. Although there is undoubtedly a story of discoveries within the mathematical world, such as the prime numbers distribution, it is also an inevitable fact that most of the mathematical terms and notations were invented by humans.

2 Examples are Powerful, but a Counterexample is a Game Changer

Even though almost everyone agrees that math does exist, it is exceedingly hard, even for mathematicians, to define what mathematics really is. Nevertheless, this chapter will bring the conclusion from experts on definition of mathematics, in line with its debatable discovery and invention. As it is way easier to experience the joy of falling in love than to describe it, it is better to get started by experiencing the history of mathematics beforehand.

2.1 The Origin Where a Line Starts and Extends Continuously

The process of mathematical discovery began in prehistoric times, with the development of ways of counting things humans needed to quantify. At its simplest, this was done by cutting tally marks in a bone or stick. People discovered the basic principles of addition and subtraction—for example, that two things (whether pebbles, berries, or mammoths) when added to another two invariably resulted in four things. While such ideas may seem obvious to us today, they were profound insights for their time. In time, words and symbols were assigned to the numbers, and the first systems of numerals began to evolve, a means of expressing operations such as acquisition of additional items, or depletion of a stock, the basic operations of arithmetic.

As hunter-gatherers turned to trade and farming, and societies became more sophisticated, arithmetical operations and a numeral system became essential tools in all kinds of transactions. To enable trade, stocktaking, and taxes in uncountable goods such as oil, flour, or plots of land, systems of measurement were developed, putting a numerical value on dimensions such as weight and length. Calculations also became more complex, developing the concepts of multiplication and division from addition and subtraction—allowing the area of land to be calculated, for example.

In early civilizations, these new discoveries in mathematics, and specifically the measurement of objects in space, became the foundation of the field of geometry, knowledge that could be used in building and toolmaking. In using these measurements for practical purposes, people found that certain patterns were emerging, which could in turn prove useful. As new applications for these mathematical discoveries were found in astronomy, navigation, engineering, bookkeeping, taxation, and so on—further patterns and ideas emerged. The ancient civilizations each established the foundations of mathematics through this interdependent process of application and discovery, while also developed a fascination with mathematics for its own sake, so-called pure mathematics. From the middle of the first millennium BCE, the first pure mathematicians began to appear in Greece, and slightly later in India and China, building on the legacy of the practical pioneers of the subject—the engineers, astronomers, and explorers of earlier civilizations.

Although these early mathematicians were not so concerned with the practical applications of their discoveries, they did not restrict their studies to mathematics alone. In their exploration of the properties of numbers, shapes, and processes, they discovered universal rules and patterns that raised metaphysical questions about the nature of the cosmos, and even suggested that these patterns had mystical properties. Often mathematics was therefore seen as a complementary discipline to philosophy, and the links between the two subjects have persisted to the present day.[3]

2.2 Partition of a Set into Disjoint Union of Its Subsets

As it evolved through insights of mathematicians from the ancient civilizations to Renaissance Europe and into the modern world, mathematics was also seen to comprise several distinct but interconnected fields of study.

The first field to emerge is the study of numbers and quantities, which we now call arithmetic, from the Greek word *arithmos* (“number”). At its most basic, it is concerned with counting and assigning numerical values to things, but also the operations, such as addition, subtraction, multiplication, and division, that can be applied to numbers.

Another major field in mathematics is algebra, which is the study of structure and the way that mathematics is organized. In its basic form, algebra is the study of the



Figure 3: Photo by Osama Elsayed on Unsplash

underlying rules of how symbols, such as letters, are used in mathematics. Methods of solving equations had been discovered as early as the ancient Babylonians, but it was medieval mathematicians of the Islamic Golden Age who pioneered the use of symbols to simplify the process, giving us the word “algebra,” which is derived from the Arabic *al-jabr*. More recent developments in algebra have extended the idea of abstraction into the study of algebraic structure, known as abstract algebra.

A third major field of mathematics, geometry, is concerned with the concept of space, and the relationships of objects in space: the study of the shape, size, and position of figures. It evolved from the very practical business of engineering and astronomy. A particular branch of geometry, trigonometry (the study of the properties of triangles), proved to be especially useful in these pursuits.

The legacy of great mathematical philosophers such as Pythagoras, Plato, and Aristotle was consolidated by Euclid, whose principles of geometry and logic were accepted as the subject’s foundation for some 2,000 years. In the 1800s, however, alternatives to classical Euclidean geometry were proposed, opening up new areas of study, including topology, which examines the nature and properties not only of objects in space, but of space itself.

Since the Classical period, mathematics had been concerned with static situations, or how things are at any given moment, but not with continuous change. Calculus, developed independently by Gottfried Leibniz and Isaac Newton in the 1600s, provided an answer to this problem. The two branches of calculus, integral and differential, offered a method of analyzing such things as the slope of curves on a graph and the area beneath them as a way of describing and calculating change.

The discovery of calculus opened up a field of analysis that later became particularly relevant to, for example, the theories of quantum mechanics and chaos theory in the 1900s.

Another field of mathematics emerged in the late 19th and early 20th centuries—the foundations of mathematics. This revived the link between philosophy and mathematics. Scholars including Gottlob Frege and Bertrand Russell inspired a re-examination of the nature of mathematics itself, how it works, and what its limits are. This study of basic mathematical concepts is an essential adjunct to every other field of modern mathematics.[3]

2.3 One-to-One Correspondence between Two Sets

Now that a variety of examples and the history of mathematics were prescribed in the previous sections, the definition of mathematics is ready to be examined prominently.

According to Oxford Languages, mathematics is the abstract science of number, quantity, and space, either as abstract concepts (pure mathematics), or as applied to other disciplines such as physics and engineering (applied mathematics).[5]

Albeit the above definition does not really cover the entire mathematical realm, it surely gives a decent picture of how mathematics looks like. It does revolve around numbers and quantities as in number theory and arithmetic, and it concerns space as in geometry. Two sides of mathematics are also mentioned: pure mathematics and applied mathematics.

Simply put, pure mathematics is about discovering any “if A , then B ” statements: Assuming something is true, what else can follow from it? This proposition is the same as saying “ A implies B ,” hence the name ‘implication’. When A and B are specified, they are called logical constants; and when A and B can be anything, they are called variables.[4]

Pure mathematics is the study of mathematics for its own sake; it revolves around concepts that are derived solely from the logical consequences of basic principles agreed upon. But applications for mathematical discoveries have usually been found, and advances in science and technology have driven innovations in mathematical thinking. And that is when applied mathematics comes into play, hence the name. While pure mathematics is based on logical constants, there are numerous instances in the real world which are not logical,—e.g. humans’ behavior or unforeseen phenomena— and so applied mathematics deals with the occurrence of non-logical constants.

In other words, pure mathematics is where one uses the world to explore mathematics; while applied mathematics is where one uses mathematics to explore the world. And fortunately, these two styles of math share a symbiotic relationship with each other.

A prime example is the connection between mathematics and computers. Originally developed as a mechanical means of doing the “donkey work” of calculation to provide tables for mathematicians, astronomers and so on, the actual construction of computers required new mathematical thinking. It was mathematicians, as much as engineers, who provided the means of building mechanical, and then electronic computing devices, which in turn could be used as tools in the discovery of new mathematical ideas. No doubt, new applications for mathematical theorems will be found in the future, too—and with numerous problems still unsolved, it seems that there is no end to the mathematical discoveries to be made.

The notion of mathematics being an “abstract science” in the definition is the remaining hole to fill in. This is because it gives a fairly vague idea of whether mathematics is a science or an art. This problem will be elaborated further in the last chapter.

3 A Proof Makes Theorem Breathe, And a Conjecture Keeps Mathematics Alive

On to the central question of the paper: Is mathematics a natural science, a branch of the humanities, or an art form? This is not necessarily an obvious thing to ask. Since the very era of our ancestors, humans love to group things together, whether it is about



Figure 4: Photo by Jexo on Unsplash

animals, social status, stars, or even area of studies, with firmly delineated boundaries. However, the nature is not so picky! This last chapter will elaborate the fuzziness of the borders between science and art when it comes to categorizing mathematics. As Albert Einstein once quoted, “As far as the laws of mathematics refer to reality, they are not certain; and as far as they are certain, they do not refer to reality”. [6]

3.1 The Derivative that Simply Describes the Rate of Change

Science has allowed us to send people to the moon, to cure tuberculosis, to invent the atom bomb, the motor car, the aeroplane, television, computers, and numerous other devices which have changed the nature of our everyday life. Scientific method is generally recognized as the most effective way we have of finding out about and predicting the behaviour of the natural world. [7]

A simple, but widespread, view of scientific method is as follows. The scientist begins by making a large number of observations of some aspect of the world: for instance, the effect of heating water. These observations should be as objective as possible: the scientist aims to be unbiased and unprejudiced in recording data. Once the scientist has gathered a large amount of data based on observations, the next stage is to create a theory which explains the pattern of results. This theory, if it is a good one, will both explain what was happening, and predict what is likely to happen in the future. If future results do not quite fit with these predictions, then the scientist will usually modify the theory to cope with them. Because there is a great deal of regularity in the natural world, scientific predictions can be very accurate.

The idea of science is, as expected, intertwined with mathematics. There is, though, a major difference between them: science proves theories through repeated experiments, while mathematics proves theorems through verified derivations in each step. Scientific theories and laws might change as reality distorts; but mathematics remains the same and is only dependent on the formerly established axioms.

Nevertheless, the usefulness of scientific method in the real world corresponds to the goal of applied mathematics: understanding reality based on mathematical reasoning. When mathematics is said to be part of science, it actually refers to the utility of its applied form. Essentially, applied mathematics is the application of mathematical methods by different fields such as physics, engineering, medicine, biology, finance, business, computer science, and industry. This also suggests that applied mathematics is hardly

distinguishable from science due to their similarities.

3.2 Recursion that Unfolds the Beauty of Fractals

Now, what about the connection between mathematics and art?

Most people who visit art galleries, read novels and poetry, watch plays or ballet, go to see films, or listen to music have at some time wondered what art is. The fact that new art forms such as film and photography have emerged and that art galleries have exhibited such things as a pile of bricks or a stack of cardboard boxes has forced us to think about the limits of what we are prepared to call art. Obviously art has meant different things to different cultures at different times: it has served ritualistic, religious, and entertainment purposes as well as embodying the beliefs, fears, and desires most central to the culture in which it was produced. In earlier times, what counted as art seemed to be more clearly defined. Yet we now seem to have reached a stage when anything whatsoever can be a work of art.[7]

There is an immense variety among works of art: paintings, plays, films, novels, pieces of music, and dance may seem to have very little in common. However, the amazing ability of mankind to have such a powerful aesthetic sense gives an intuition that all of them are linked in a beautiful manner that is exceedingly difficult to describe.

Surprisingly, the idea of pure mathematics is somewhat aligned with the concept of art as well. People are engaged with pure mathematics not because of their usefulness, but because of the plain beauty and elegance of the proofs and theorems within it. Likewise, artists are attached to art pieces simply because they adore their charm. It is hard to imagine that someone would watch a romance movie just because she would use it in her life!

The fact that there are two sides of mathematics smoothly urges that it also has both a scientific nature and an artistic sense in it. For some people, it is surprising to realize that humans' intuition to separate two different ideas is often denied by the will of nature that sets a fuzzy barrier between science and art. To make it more convincing, it is not just happening in mathematics; the philosophy of architecture also shares a similar manner!

3.3 The Proof Left as an Exercise to the Reader

Mathematics puts people in their place, indeed. What makes it unique is that all the results and facts follow smoothly from predefined notions. It reflects the face of art in its beauty, but also the face of science in its utility. Therefore, the following simple sentence concludes this paper's objective of answering the question: What exactly is mathematics?

*Mathematics is the axiomatic set of symbolic logic that expresses
both art in its purity and science in its applied form.*

The story of mathematics is one of exploration of these different fields, and the discovery of new ones. But it is also the story of the explorers, the mathematicians who set out with a definite aim in mind, to find answers to unsolved problems, or to travel into unknown territory in search of new ideas—and those who simply stumbled upon an idea in the course of their mathematical journey, and were inspired to see where it would lead.

Sometimes the discovery would come as a game-changing revelation, providing a way into unexplored fields; at other times it was a case of “standing on the shoulders of giants,”



Figure 5: Photo by Wojciech Then on Unsplash

developing the ideas of previous thinkers, or finding practical applications for them. For instance, who would have thought that the infiniteness of prime numbers actually made its way to its usefulness in the digital world of secret communication called cryptography!

This paper unavoidably still leaves many questions about the true meaning of mathematics. However, it is actually intended to be so, and those unanswered questions are left for fellow readers to ponder about, because that is how philosophers and mathematicians work. Philosophy is still alive because people never stop asking questions, and mathematics is still alive because conjectures are still being made up until today; and that is what makes it exhilarating!

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